Model 1 Writing Brief

Coupled vegetation–water–sand dynamics (Task 1)

Source of every number: `Results/v7/model1/` (run of `Code/v7_model1.py`)

2026-11-11

This brief collects what you need to write the Model 1 section of the paper: the goal, how the model is built, every formula with its meaning, parameter values, the results with numbers, and what the results do and do not allow you to claim. All values are copied from the result tables; figure files are in `Figures/M1/`. Units: V is in percentage points of a cover proxy, M in mm, irrigation in mm/yr on the treated area (1 mm = 10 m^3/ha).

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1 Goal and what we achieved

Task 1 asks for a coupled dynamic model of vegetation cover and effective soil moisture on a representative section of the desert margin, using real data, and for the factors that drive them.

Requirement	How Model 1 meets it	Evidence
Representative section (width tens of m to km, length tens to hundreds of km)	Yutian–Minfeng southern margin of the Taklimakan, centre line through windows CON1–CON4, length 127.95 km, assumed width 1 km	`section_definition.json`
Coupled *dynamic* model	Two-state difference system: yearly cover V_t and within-year storage M_j, solved together every year	Eqs. (1)–(5)
State variables V and M	V: growing-season NDVI cover proxy; M: plant-available water above wilting point (mm)	Section 3
Use data; find influencing factors	MODIS NDVI 2015–2025 at 7 stations, daily wind, monthly P, ET_0, T; influence of irrigation, rain, demand, efficiency, wind quantified	Tables 4, 6
Water–vegetation–sand feedback	Water → growth ($\overline{W}$); cover → water demand; cover → wind shading; wind → cover loss	Fig. 1

Table 1: Task 1 requirements and how they are met.

Main results in one paragraph. Pooled over 7 oasis stations, the intrinsic growth rate is $r = 0.547$ /yr and the decay rate $d = 0.107$ /yr; cover would halve in about 6.1 years without water. Under 2015–2022 climate (52 mm/yr rain against 1,686 mm/yr reference evapotranspiration), the long-run cover proxy rises from 1.8% with no irrigation to 17.9% at 300 mm/yr and 24.1% at 600 mm/yr, with sharply falling returns, and saturates at 30.6% above about 1,500 mm/yr. Stopping irrigation after 10 years drops cover from 21.5% to 5.1% within 15 years. The one-year-ahead forecast does **not** beat "next year equals this year" on held-out years (RMSE 0.0635 vs 0.0536), so the model is a scenario tool, not a forecaster.

2 Model construction

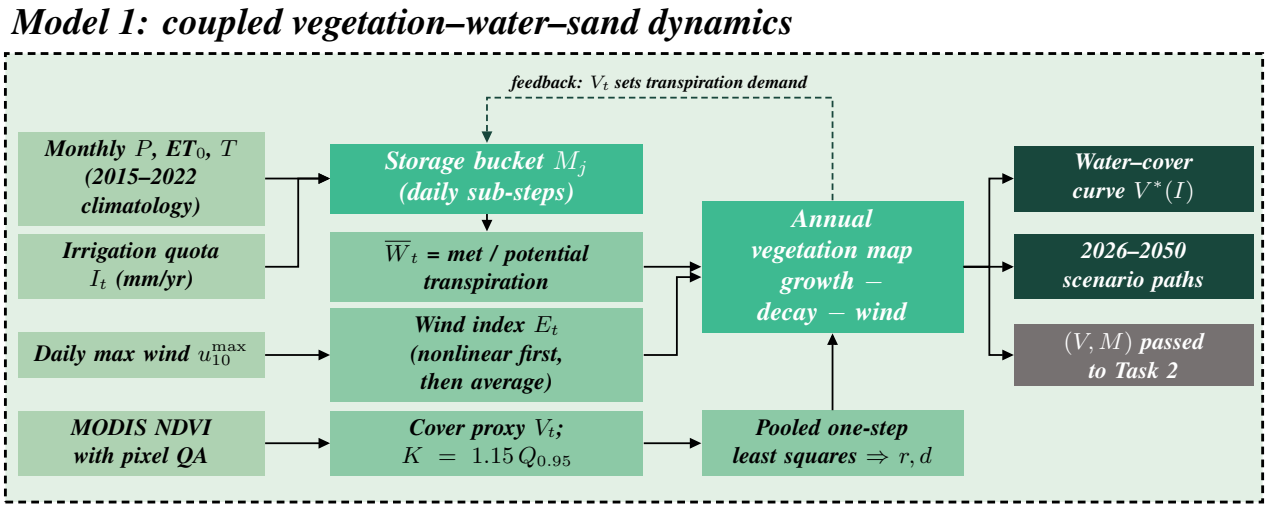


Figure 1: Structure of Model 1. Inputs on the left; the storage bucket runs on daily sub-steps inside each year; the vegetation map updates once per year; outputs on the right.

Two states, two time scales.

- V_t (yearly): mean NDVI-derived cover proxy of the June–September growing season of year t, with end-members NDVI 0.05 (bare) and 0.85 (full): $V = 100\,\mathrm{clip}\big((\mathrm{NDVI} - 0.05)/0.80, 0, 1\big)$.

- M_j (daily sub-steps j within each year, carried across years): water above wilting point in the root zone, $0 \le M_j \le C$.

Order of computation in each year. (1) Hold V_t fixed; (2) run the storage bucket over every day of the year with that year's rain, ET_0 and irrigation; (3) sum transpiration into the met-demand ratio $\overline{W}_t$; (4) update V_{t+1} with $\overline{W}_t$ and the wind index E_t; (5) carry the year-end storage to next year.

Three processes in the vegetation equation. Logistic growth limited by water and a local ceiling K; first-order natural decay; wind damage that weakens as cover increases (shading).

Four feedback links. water → vegetation (growth is multiplied by $\overline{W}$); vegetation → water (more cover means more transpiration demand, so lower $\overline{W}$); vegetation → sand (shading factor $1 - \eta V/K$); sand → vegetation (wind damage $kE_{\text{eff}}V$).

Writing note: Present Fig. 1 first, then the vegetation equation, then the bucket. Stress that $\overline{W}$ is a transpiration ratio, not soil moisture; M is the soil-water state.

3 Formulas and what they mean

3.1 Vegetation update (yearly)

$$V_{t+1} = V_t + \underbrace{rV_t\left(1 - \frac{V_t}{K}\right)\overline{W}_t}_{\text{growth}} - \underbrace{dV_t}_{\text{natural decay}} - \underbrace{kE_t\left(1 - \eta\frac{V_t}{K}\right)V_t}_{\text{wind damage}} \tag{1}$$

- **Growth.** rV is exponential growth; $(1 - V/K)$ stops growth at the local ceiling K; $\overline{W}_t \in [0, 1]$ scales growth by how much of the plants' water demand was met. Temperature is already inside r (one yearly growth season), so no extra temperature factor is multiplied here.
- **Decay.** A fixed share d of cover dies each year. With no water and no wind, $V_{t+1} = (1 - d)V_t$, so the half-life is $\ln(1/2)/\ln(1 - d)$ years.
- **Wind damage.** Loss proportional to cover and to the wind index E_t; the factor $(1 - \eta V/K)$ means denser cover shelters itself ($\eta = 1$: full shelter at $V = K$).
- $V = 0$ stays at 0: water alone cannot create plants; new stands need a planted initial cover.

3.2 Effective soil-water storage (daily sub-steps)

Monthly rain P, ET_0 and the month's share of irrigation are spread evenly over the days of that month. Potential transpiration on day j:

$$D_j = c\ \min(1, 2f_{c,t})\ f(T_j)\ \mathrm{ET}_{0,j}, \qquad f_{c,t} = \mathrm{clip}(V_t/V_{\text{full}}, 0, 1). \tag{2}$$

Demand grows with cover until V reaches half of V_{full}, and follows the season through $f(T)$. Yearly irrigation I_t is split over months in proportion to $f(T)\mathrm{ET}_0$ (watering follows demand).

$$B_j = M_j + \alpha_P P_j + \eta_w I_j + G_j \qquad \text{water available today} \tag{3}$$

$$Q_j = \max(B_j - C, 0) \qquad \text{overflow to deep drainage}$$

$$K_{s,j} = \min\{1,\ (B_j - Q_j)/[(1 - p)C]\} \qquad \text{FAO-56 stress coefficient}$$

$$T_{a,j} = \min(B_j - Q_j,\ K_{s,j}D_j) \qquad \text{actual transpiration}$$

$$M_{j+1} = B_j - Q_j - T_{a,j} \qquad \text{storage carried to next day} \tag{4}$$

$$\overline{W}_t = \textstyle\sum_{j\in t} T_{a,j} \big/ \sum_{j\in t} D_j \qquad \text{share of demand met in year } t \tag{5}$$

Every step conserves water exactly: $M_{j+1} - M_j = \alpha_P P_j + \eta_w I_j + G_j - T_{a,j} - Q_j$.

- α_P and η_w are the shares of rain and of gross irrigation that reach the roots; interception, soil evaporation and conveyance losses are inside them.
- Plants are unstressed while storage is above $(1-p)C$; below that, transpiration falls linearly.
- When storage is full, extra water drains (Q_j) and does not raise $\overline{W}$.

3.3 Helper relations

$$f(T) = \min\{1, \max\{0, (T-5)/20\}\} \qquad \text{0 below 5 °C, 1 above 25 °C} \tag{6}$$

$$q_i = u_i^2 \max(u_i - u_T, 0), \quad u_T = 5 \text{ m/s} \qquad \text{daily sand-drift proxy from daily max wind} \tag{7}$$

$$E_t = \frac{\frac{1}{n_t}\sum_{i \in t} q_i / \bar{q}}{\text{training-period mean of the same}} \qquad \text{wind index, 1 = training mean} \tag{8}$$

$$K = 1.15 \times Q_{0.95}(V_{\text{train}}) \qquad \text{local ceiling from the 95th percentile} \tag{9}$$

The wind proxy is computed on each day and averaged afterwards; averaging the wind first would hide most gusts above the 5 m/s threshold and make E almost zero.

3.4 Long-run state and the closed form

With $x = V/K$ and $\overline{W}$, E held fixed, the non-zero steady state is

$$x^* = \frac{r\overline{W} - d - kE}{r\overline{W} - kE\eta}, \qquad \text{exists if } r\overline{W} > d + kE. \tag{10}$$

Because $\overline{W}$ itself depends on V, the real long-run state is the fixed point of the yearly map $(V, M) \mapsto (V', M')$, found by iterating from a low and a high start until both agree (residual $< 10^{-7}$). With ample water ($\overline{W} = 1$) and $k = 0$: $x^* = 1 - d/r = 0.804$, i.e. $V^* = 30.6\%$ at $K = 38.08\%$, which is exactly the saturation value found numerically.

3.5 Calibration and checks

$$\min_{r,d} \frac{1}{N} \sum_{(s,t) \in \text{train}} \left(\tilde{x}_{s,t}\left[1 + r(1 - \tilde{x}_{s,t}) - d\right] - \tilde{x}_{s,t+1} \right)^2 \tag{11}$$

One-step transitions at 7 irrigated oasis stations, pooled, assuming $\overline{W} = 1$ and $k = 0$ there; training = target year ≤ 2022 (49 transitions), hold-out = 2023–2025 (21 transitions). The residual is linear in (r, d), so the least-squares solution is unique. Uncertainty: local covariance, leave-one-station-out, and 300 whole-station bootstrap resamples. Skill: RMSE in $x = V/K$, compared with persistence ($\hat{x}_{t+1} = x_t$) and the training-period station mean.

Writing note: Explain why r, d are fitted only at oases: there $\overline{W} \approx 1$, so growth and decay can be separated from water. If you fit free r, d, K on a planted section, only $r - d$ and r/K are identifiable.

4 Parameters

Symbol	Value (range tested)	Unit	Meaning and status
r	0.547 (bootstrap 95%: 0.335–1.168)	yr^{-1}	intrinsic growth rate; fitted
d	0.107 (bootstrap 95%: 0.077–0.202)	yr^{-1}	natural decay rate; fitted
K	38.08 (section); per station $1.15Q_{0.95}$	%	cover ceiling; rule from data
k	0 (0, 0.02, 0.05, 0.10)	–	wind damage; scenario (fit not identifiable)
η	1	–	shading strength; scenario
C	75 (30–150)	mm	storage capacity; scenario
p	0.5 (0.3–0.7)	–	stress-free share of storage (FAO-56); scenario
c	1.0 (0.7–1.2)	–	transpiration coefficient; scenario
η_w	0.75 (0.65–0.85)	–	share of gross irrigation that reaches roots; scenario
α_P	0.25 (0–1)	–	share of rain that reaches roots; scenario
G	0	mm/yr	groundwater supply; scenario
V_{full}	66.59 (64.85–69.40)	%	full-cover end-member (local oasis NDVI 0.5827)
u_T	5	m/s	wind threshold for sand drift
V_0, M_0	2, 37.5	%, mm	initial state of a new stand in scenarios

Table 2: Model 1 parameters. "Scenario" means no field data to fit it; values are stated assumptions with tested ranges.

Data and forcing.

- Stations: ALE, KEL, MF, QM, RQ, YL, YT (vegetated; all 7 pass the training missing-data rule). MODIS 250 m 16-day NDVI, pixel QA 0/1 only, ≥ 50% valid pixels per composite, ≥ 2 valid growing-season months per year. 395 of 396 growing-season station-months valid; no interpolated values are used for fitting or scoring.
- Climate for scenarios: 2015–2022 monthly climatology of Yutian/Minfeng (Open-Meteo archive): rain 52.2 mm/yr, ET_0 1,685.6 mm/yr, temperature-weighted ET_0 1,190.8 mm/yr; wind index peaks in April (2.29) and May (2.08).
- Raw MODIS replay: all 198 downloaded chunks match the original SHA-256 hashes.

5 Results

5.1 Calibration and model selection

Model	Parameters	Cond. number	AIC	RMSE train	RMSE hold-out	R^2 hold-out
M0 (r, d) **adopted**	$r = 0.547 \pm 0.174$, $d = 0.107 \pm 0.033$	375	−286.48	0.0516	0.0635	0.463
M1 $(+k)$	$r = 0.778$, $d = 0.105$, $k = 0.249 \pm 0.431$	4,406	−284.83	0.0514	0.0657	0.424
M2 $(+k, \eta)$	same, $\eta = 1.0$ at bound	94,351	−282.83	0.0514	0.0657	0.424
Persistence	$\hat{x}_{t+1} = x_t$	–	–	0.0572	**0.0536**	–
Station mean	training mean	–	–	–	0.0737	–

Table 3: Nested fits (units of $x = V/K$). Adding wind damage worsens AIC and hold-out error and the uncertainty of k exceeds its value, so k is kept as a scenario parameter.

- Hold-out skill by station (MSE skill vs persistence): MF +0.44, YT +0.20; ALE −0.44, RQ −0.52, YL −0.57, KEL −2.62, QM −2.88.
- Leave-one-station-out: r stays in 0.47–0.53 and d in 0.09–0.11 for six stations; leaving out YT gives $r = 0.958$, $d = 0.168$ (YT is influential).

- Bootstrap (300 resamples of whole stations, only 7 clusters): r median 0.554, 95% 0.335–1.168; d median 0.109, 95% 0.077–0.202.
- Stricter QA (75% valid pixels, 6 stations): $r = 0.510$, $d = 0.103$; conclusions unchanged.
- Free run from 2022 to 2025: the model stays near 2022 values while most stations greened in 2023–2025 (e.g. RQ predicted 33.0% vs observed 38.4% in 2025; QM 30.7% vs 37.2%).
- Section pixels (retrospective): station parameters transferred to planted pixels of CON1–3 give $R^2 = 0.398$ (RMSE 8.1 points, $n = 116$); CON4 gives $R^2 = 0.318$ (RMSE 12.1 points, $n = 129$). A free fit there gives net rate $r - d = 0.543$ and plateau 35.3%.
- Water bookkeeping: largest balance error over all scenarios 5.3×10^{-13} mm. No-water half-life: 6.1 years.

5.2 Irrigation–cover curve (long-run state)

Irrigation (mm/yr)	m^3/ha/yr	V^* (%)	M^* end of year (mm)	$\overline{W}^*$	Relative erosion $1 - \eta V^*/K$
0	0	1.77	7.1	0.21	0.95
50	500	6.05	8.3	0.23	0.84
100	1,000	9.43	9.4	0.26	0.75
150	1,500	12.17	10.5	0.29	0.68
200	2,000	14.43	11.5	0.32	0.62
300	3,000	17.94	13.4	0.37	0.53
400	4,000	20.54	15.4	0.43	0.46
500	5,000	22.54	17.3	0.48	0.41
600	6,000	24.14	19.3	0.54	0.37
800	8,000	26.51	23.2	0.65	0.30
1,000	10,000	28.20	27.0	0.76	0.26
1,500	15,000	30.60	72.8	1.00	0.20
2,000	20,000	30.60	74.4	1.00	0.20

Table 4: Long-run state for constant irrigation (fixed point, converged from both starts). The last column is the shading factor, not a measured erosion reduction.

- **Diminishing returns.** The first 100 mm/yr adds 7.7 points of cover (0.077 points per mm); 500→600 mm adds 1.6 points (0.016 per mm); 800→1,000 mm adds 1.7 points (0.008 per mm).
- **Saturation.** Above about 1,500 mm/yr, $\overline{W} = 1$, storage is full and the extra water drains; cover is capped at $K(1 - d/r) = 30.6\%$.
- **Rain alone** (52 mm/yr) holds only 1.8% cover.

5.3 Scenario paths 2026–2050

Start: $V_0 = 2\%$, $M_0 = 37.5$ mm; climate repeats the 2015–2022 average year (not a weather forecast).

Scenario	2030	2035	2036	2040	2050
No irrigation	2.19	1.99	1.97	1.89	1.80
100 mm/yr	6.25	7.99	8.20	8.78	9.30
300 mm/yr	9.90	15.50	16.00	17.16	17.86
600 mm/yr	9.90	21.53	22.31	23.69	24.13
1,000 mm/yr	9.90	25.15	26.43	27.99	28.20
600 mm/yr, stopped after 2035	9.90	21.53	19.46	12.74	5.09

Table 5: Cover proxy V (%) at the end of each year.

- **Establishment is growth-limited.** Every quota of 300 mm/yr or more gives the same 9.90% in 2030: young stands use little water, storage stays full and the surplus drains. Water above ~300 mm/yr is wasted in the first five years.
- **Stopping water undoes the work.** After irrigation stops, $\overline{W}$ falls to 0.03–0.07 and cover drops from 21.5% to 12.7% in 5 years and to 5.1% in 15 years.
- It takes about 15–25 years to get close to the long-run value (300 mm/yr: 17.86% in 2050 vs 17.94% long-run).

5.4 Sensitivity

Parameter	Low / high	V^* at 300 mm/yr		V^* at 600 mm/yr	
		low	high	low	high
Baseline	–	17.94		24.14	
Transpiration coefficient c	0.7 / 1.2	21.32	16.22	27.12	22.49
Irrigation efficiency η_w	0.65 / 0.85	16.66	19.06	22.89	25.20
Rain share α_P	0 / 1	17.40	19.38	23.89	24.85
Full-cover end-member V_{full}	64.85 / 69.40	17.68	18.33	23.91	24.50
Storage capacity C	30 / 150 mm	17.94	17.94	24.14	24.14
Stress share p	0.3 / 0.7	17.94	17.94	24.14	24.14
Wind damage k (300 mm/yr)	0 / 0.02 / 0.05 / 0.10	17.94 / 17.01 / 15.66 / 13.62			

Table 6: Long-run cover under one-at-a-time changes.

- The demand side (c) and the efficiency of water delivery (η_w) matter most; each moves V^* by 1–3 points.
- Storage capacity and p do not change the long-run cover (they change M only). Reason: in a repeating year without overflow, storage returns to the same value, so yearly transpiration equals yearly effective inflow and $\overline{W}$ depends only on inflow and demand. Capacity matters for the timing within the year and for the start-up phase.
- Wind damage is the largest single uncertainty: $k = 0.10$ removes 24% of the cover at 300 mm/yr.

6 What we can claim and what we cannot

Can claim	Must not claim
A mass-conserving coupled V–M model, calibrated on real NDVI with stated assumptions.	That the water response is field-calibrated (no measured irrigation–storage–cover data).
Growth and decay rates with honest, wide uncertainty.	That the model beats persistence for one-year forecasts (it does not).
Irrigation has strongly diminishing returns; establishment needs little water; stopping water causes collapse within ~15 yr.	Measured ground cover: V is an NDVI proxy; 1 km width and 128 km length are a study scenario, not an engineering boundary.
Wind damage could remove up to a quarter of the cover (scenario).	Erosion reduction in tonnes or percent; CON4 as an independent validation (pixels were selected using later greening).

7 Figures ready for the paper

File (`Figures/M1/`)	What it shows / caption point
`m1_f01_station_series`	7 stations 2015–2025: observed V, one-step prediction, free run from 2022, persistence, local K
`m1_f02_holdout_skill`	Observed vs predicted; per-station hold-out RMSE vs persistence (negative skill kept)
`m1_f03_parameter_bootstrap`	r–d cloud from bootstrap, leave-one-out points, pooled fit
`m1_f04_cohort_diagnostic`	Planted pixels CON1–4 vs transferred curve (retrospective only)
`m1_f05_forcing_climatology`	Monthly rain, ET_0, temperature-weighted ET_0, wind index
`m1_f06_water_cover_curve`	Irrigation vs long-run V, $\overline{W}$, M, relative erosion
`m1_f07_scenario_trajectories`	2026–2050 paths for constant quotas and the stop-after-10 case
`m1_f08_monthly_water_balance`	Monthly inflow, transpiration, drainage and storage
`m1_f09_sensitivity`	Tornado chart of water parameters; wind damage scenarios

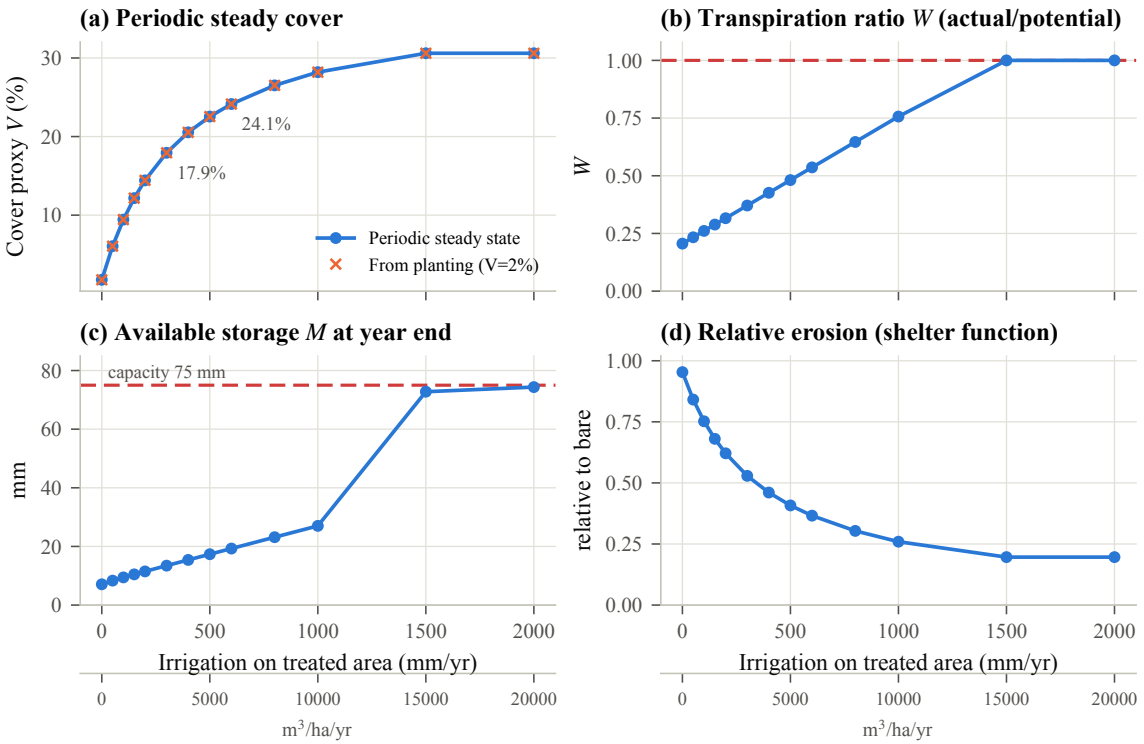


Figure 2: (From `m1_f06`) Irrigation–cover curve. Use it for the "diminishing returns" and "saturation" points.

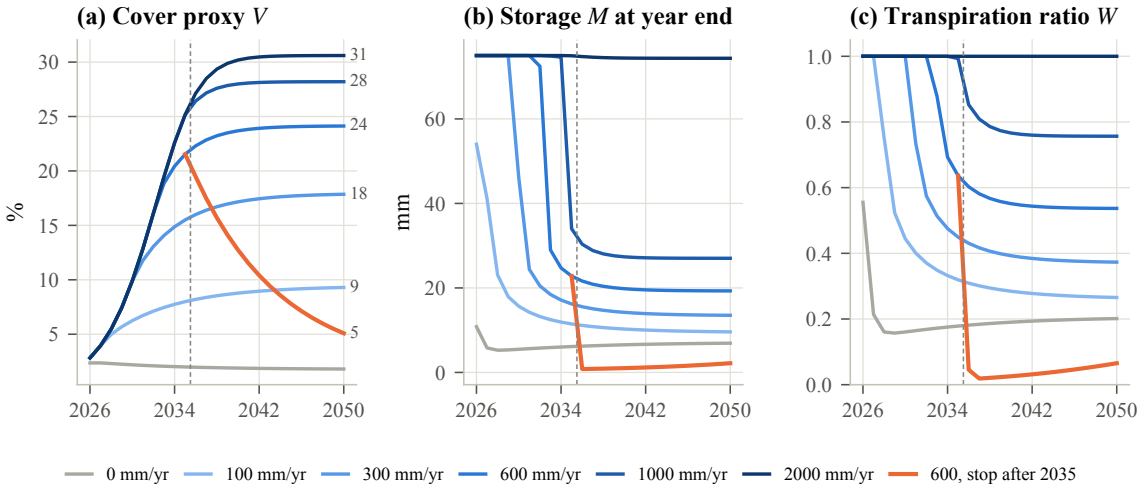


Figure 3: (From `m1_f07`) Scenario paths 2026–2050, including stopping irrigation after 10 years.

8 Suggested outline for the Model 1 section

1. Section choice and data (Yutian–Minfeng, 127.95 km × 1 km scenario; stations; QA rules).
2. Model structure (Fig. 1) and the four feedback links.
3. Equations (1)–(5) with one line each on meaning.
4. Calibration (Eq. (11)), Table 3, honest skill statement, bootstrap.
5. Results: irrigation–cover curve (Table 4), scenario paths (Table 5).
6. Influencing factors: sensitivity (Table 6), wind damage scenarios.
7. Limits and the hand-over to Model 2: each planted cohort carries its own (V, M).